

Chapter 17:
Communication and Internet technologies

Learning objectives

By the end of this chapter you should be able to:

- show understanding of circuit switching
- show understanding of packet switching
- show understanding of why a protocol is essential for communication between computers
- show understanding of how protocol implementation can be viewed as a stack, where each layer has its own functionality
- show understanding of the TCP/IP protocol suite
- show understanding of protocols (HTTP, FTP, POP3, IMAP, SMTP, BitTorrent) and their purposes.



17.01 Transmission modes

For communication over an internetwork there are two possible approaches: circuit switching or packet switching.

Circuit switching

Circuit switching is the method used in the traditional telephone system. Because the Public Switched Telephone Networks (PTSNs) have now largely converted to digital technology, the same method can be provided for data transfer that was traditionally used for voice communication. In Chapter 2 (Section 2.06) the concept of support for Internet usage by a PTSN was introduced. Typically, this is provided in a leased line service. The concept is illustrated in Figure 17.01, which shows end-systems connected to local exchanges which have a switching function and which are connected via a number of intermediate nodes with a switching function.

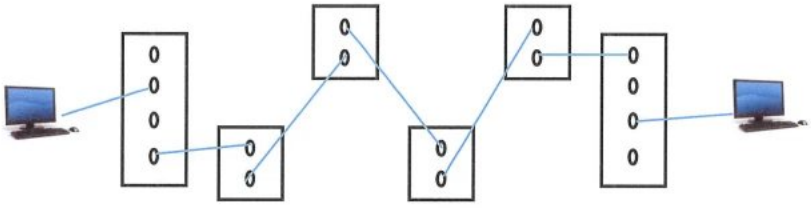


Figure 17.01 Circuit-switched data transmission

For data transfer to take place, the following has to happen.

1. The sender provides the identity of the intended receiver.
2. The system checks whether or not the receiver is ready to accept data.
3. If the receiver is available, a sequence of links is established across the network.
4. The data is transferred.
5. The links are removed.

It is not necessary for this discussion to define what could constitute a node in a circuit-switched network. The links that are provided between the nodes are dedicated channels in shared transmission media that guarantee unimpeded transmission. When a telephone call has finished there is a definite end to the call with removal of the links. However, for a leased-line data connection there might be a permanent circuit established.

Packet switching

The packet-switching method allows data transmission without a circuit being established. Data cannot be sent in a continuous stream. Instead data is packaged in portions inside packets. A packet consists of a header which contains instructions for delivery plus the data body. The method is similar to that used by the postal service but rather more complex! The network schematic shown in Figure 17.01 is still appropriate to describe packet switching except that the links used are not defined at the time a packet is transmitted by the sender. Furthermore, the nodes will have an extended functionality compared to that required in a circuit-switched transmission. How a router acts as a node and supports packet switching is discussed in Section 17.04.

When packet switching is used, there are two ways that the network can provide a service: connectionless service or connection-oriented service. If a connectionless service is provided, a packet is dispatched with no knowledge of whether or not the receiver is ready to accept the packet, and has no way of finding out if the transmission has

succeeded. In a connection-oriented service the first packet sent includes a request for an acknowledgement. If the acknowledgement is received, the sender transmits further packets. If no acknowledgement is received, the sender tries again with the first packet.

17.02 Protocols

The basic definition of a **protocol** is simple – it is a set of rules. So, what do these rules relate to? Before answering, it should be understood that we often talk about 'a protocol' when we are referring to a protocol suite which contains more than one individual protocol. The complexity of networking requires many individual protocols. A further complication is that there might be a number of different versions of a protocol. Finally, there are often protocols that can be used to complement the use of another protocol.

KEY TERM

Protocol: a set of rules for data transmission which are agreed by sender and receiver

The set of rules that constitute a protocol must be agreed between the sender and the receiver for any communication transmitted over a network. At the simplest level, a protocol could define that a positive voltage represents a bit with value 1. A protocol might define a transmission speed that a sender must not exceed. Many of the rules relate to the format of a message or of a component of a message. For example, a definition of the format of the first 40 bytes in a packet.

17.03 A protocol stack

For a protocol suite the protocols can be viewed as layers within a protocol stack. There are a number of aspects relating to this concept.

- Each layer can only accept input from the next higher layer or the next lower layer.
- There is a defined interface between adjacent layers which constitutes the only interaction allowed between layers.
- A layer is serviced by the actions of lower layers.
- With the possible exception of the lowest layer the functioning of a layer is created by installed software.
- A layer may comprise sub-layers.
- Any user interaction will take place using protocols associated with the highest level layer in the stack.
- Any direct access to hardware is confined to the lowest layer in the stack.

17.04 The TCP/IP protocol suite

TCP/IP is the protocol suite underpinning Internet usage. TCP/IP can be explained on the basis of the network model shown in Figure 17.02. It can be seen that TCP/IP only occupies the top three layers of this model.

TIP

Unfortunately there are different names used for two of the layers. Figure 17.02 shows the Network layer and the Data link layer. In some sources these will be named as the Internet layer and the Link layer.

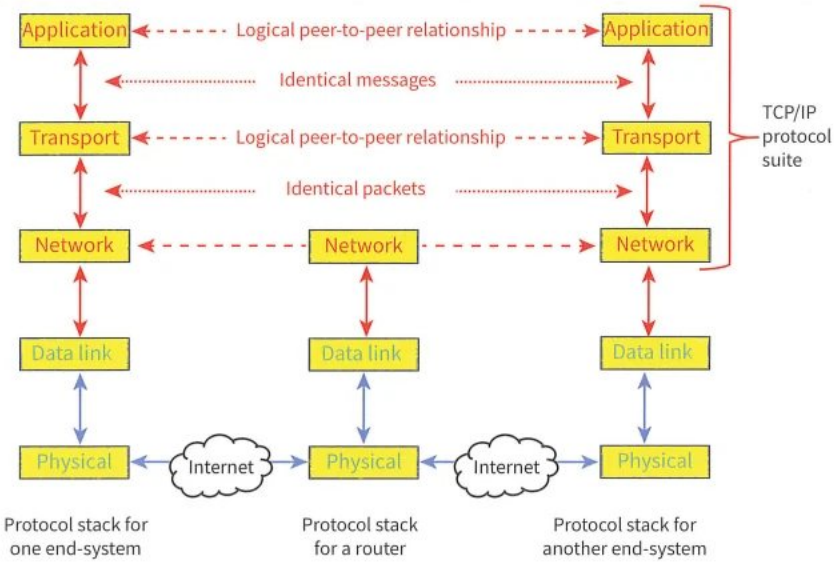


Figure 17.02 A network model relating to the TCP/IP protocol suite

There are two end-systems and the diagram shows the logical connections between corresponding layers in these two systems. An application can run on one end-system and behave as though there was a direct connection with an application running on a different end-system. The application layer protocol on the sender end-system sends a 'message' to the transport layer protocol on the same system. The transport layer protocol then initiates a process which results in the identical 'message' being delivered to the receiver end-system. On the receiver end-system, the final stage is the transport layer protocol delivering the 'message' to the application layer protocol.

The TCP/IP suite comprises a number of protocols, including the following:

- application layer: HTTP, SMTP, DNS, FTP, POP3, IMAP
- transport layer: TCP, UDP, SCTP
- network layer: IP, IGMP, ICMP, ARP.

The selection has been chosen to illustrate that the TCP/IP suite encompasses a very wide range of protocols that is still evolving. Some of the listed protocols will not be considered further.

TCP (Transmission Control Protocol)

If an application is running on an end-system where a 'message' is to be sent to a different end-system the application will be controlled by an application layer protocol as described above. The protocol will transmit the user data to the transport layer. The TCP protocol operating in the transport layer now has to take responsibility for ensuring the safe delivery of the 'message' to the receiver. The TCP protocol creates sufficient packets to hold all of the data. Each packet consists of a header plus the user data.

Question 17.01

Can you find out some details about the format of a packet that TCP would send to IP?

As well as ensuring safe delivery, TCP has to ensure that any response is directed back to the application protocol. So, one item in the header is the port number which identifies the application layer protocol. For example, for HTTP the port number is 80. The packet must also include the port number for the application layer protocol at the receiving end-system. However, TCP is not concerned with the address of the receiving end-system. If the packet is one of a sequence, a sequence number is included to ensure eventual correct reassembly of the user data.

The TCP protocol is connection-oriented. In accordance with the procedure described in Section 17.01, initially just one packet of a sequence is sent to the network layer. Once the network layer returns an acknowledgement to the Transport layer indicating that the connection has been established, TCP sends the other packets and receives response packets containing acknowledgements. This allows missing packets to be identified and re-sent.

IP (Internet Protocol)

The function of the network layer, and in particular of the IP, is to ensure correct routing over the Internet. To do this the IP protocol takes the packet received from the transport layer and adds a further header. The header contains the IP addresses of both the sender and the receiver. To find the IP address of the receiver, it is very likely to use the DNS service to find the address corresponding to the URL supplied in the user data. The DNS service is discussed in some detail in Chapter 2 (Section 2.09).

The IP packet, which is usually called a 'datagram', is sent to the data-link layer and therefore to a different protocol suite. The data-link layer assembles datagrams into 'frames' as discussed in the next section of this chapter. Once the IP packet has been sent to the data-link layer, IP has no further duty. IP functions as a connectionless service. Once a packet has been sent, IP will have no knowledge of whether or not it reached its destination. If IP receives a packet which contains an acknowledgement of a previously sent packet, it will simply pass the packet on to TCP with no awareness of the content.

The router

As Figure 17.02 shows, the frame sent by the data-link layer will arrive at a router during transmission (more likely at several routers). At this stage, the datagram content of the frame is given back to IP. It is now the function of the router software to choose the next target host in the transmission. The software has access to a routing table appropriate to that router. The size and complexity of the Internet prohibits a router from having a global routing table. Once the appropriate address has been inserted into the datagram, IP passes it back to the data-link layer of the router.

The routing table for every router has details of any current problems with any of the options for the next transmission step. This ensures that packets are delivered to their destination in the shortest possible time available.

The major distinction between a switch and a router as a node in a network is that when a frame arrives at a switch, it is transmitted on without any routing decision. A switch operates in the data-link layer but has no access to the network layer.

17.05 The Ethernet protocol stack

As discussed in Chapter 2 (Section 2.05), Ethernet is a protocol suite designed for use in a local area network (LAN). As such it can function in an isolated LAN with no connection to the Internet or any other network. However, it is now almost inevitable that a LAN will be connected to the Internet and, therefore, a LAN's protocol suite will support the protocol suite in use for the Internet.

If we look at the protocol stack for one end-system, as shown in Figure 17.02, we can see that the TCP/IP protocol suite occupies the top three layers of the five-layer stack and is therefore supported by the lower two layers. TCP/IP has no concern with the functioning of these two lower layers; it is designed to be capable of being supported by whatever protocols are available. It should be noted that some sources only use a four-layer stack for an end-system. This is either a decision to only include layers that are handled entirely by software. Or it is a decision to amalgamate all of the support for TCP/IP into one layer.

Ethernet is the most likely protocol to be used to provide the functionality required of the two lower layers. Logically the Ethernet suite can be viewed as comprising two sub-layers for each of the Data link and Physical layers. This is illustrated in Figure 17.03.

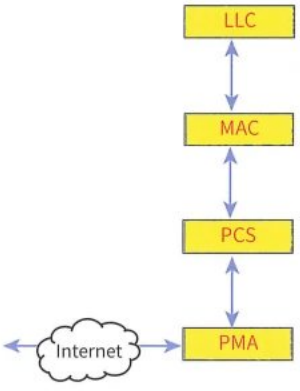


Figure 17.03 The sub-layers of Ethernet

The following points explain how Ethernet functions when supporting TCP/IP.

- The Logical Link Control (LCC) protocol is responsible for the interaction with the Network layer. It manages data transmissions and ensures the integrity of data transmissions. However, because Ethernet is a connectionless protocol it has no responsibility for checking that a transmission has been successfully delivered.
- The Medium Access Control (MAC) protocol is responsible for assembling the Ethernet packet which is referred to as a frame. Two components of this are the address of the transmitter and the address of the receiver. In addition the MAC protocol initiates frame transmission and handles recovery from transmission failure due to a collision (possibly using CSMA/CD).
- The Physical Coding Sublayer (PCS) protocol is responsible for coding data ready for transmission or decoding data received. It either receives a frame from the MAC protocol or sends one to it.
- The Physical Medium Attachment (PMA) protocol is responsible for signal transmitting and receiving.

MAC addresses

Both addresses in the Ethernet frame are examples of what are called physical or MAC addresses. A MAC address is the one which uniquely defines one NIC, as described in Chapter 2 (Section 2.04).

The reason that unique addresses have so far been guaranteed is that the 48 bits used for the definition have given a sufficient number of different addresses. However, there is a 64-bit alternative which is already used occasionally but is available for future use when 48 bits are insufficient. The 48-bit address is usually written in hexadecimal notation, for example:

4A:30:12:24:1A:10

In one version of this addressing scheme the first three bytes identify the manufacturer.

17.06 Application-layer protocols associated with TCP/IP

There are very many application-layer protocols. This discussion considers some that are used most often.

HTTP (HyperText Transfer Protocol)



TIP

Be careful not to confuse HTTP and HTML.

Because HTTP (HyperText Transfer Protocol) underpins the World Wide Web it has to be considered to be the most important application-layer protocol. Every time a user accesses a website using a browser, HTTP is used but its functionality is hidden from view.

HTTP is a transaction-oriented, client–server protocol. The transaction involves the client sending a 'request' message and the server sending back a 'response' message. The HTTP protocol defines the format of the message. The first line of a request message is the 'request line'. Optionally this can be followed by header lines. All of this uses ASCII coding. The format of the request line is:

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<Method> <URL> <Version>CRLF
```

where CR and LF are the ASCII carriage return and line feed characters. The request line usually has GET as the method. However, there are several alternatives to the GET method which makes HTTP potentially a more widely applicable protocol than just being used for webpage access. The version has to be specified because HTTP has evolved so there is more than one version in use.

Email protocols

The traditional method of sending and receiving emails is illustrated in Figure 17.04 which shows that three, individual, client–server interactions are involved. The email sender acting as a client has a connection to a mail server. This server then has to function as a client in the transmission to the mail server used by the email receiver acting as a client.

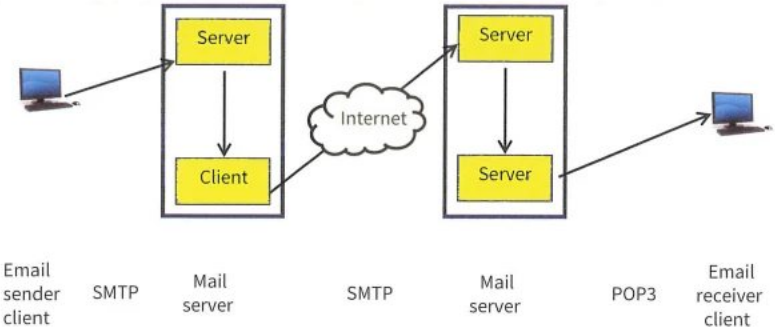


Figure 17.04 An email message being transmitted from a sender to a receiver

Of the two protocols shown being used, SMTP (Simple Mail Transfer Protocol) is a 'push' protocol whereas POP3 (Post Office Protocol version 3) is a 'pull' protocol. There is a more recent alternative to POP3, which is IMAP (Internet Message Access Protocol). The approach using POP3 is for emails to be downloaded onto the client computer. With IMAP the emails are not downloaded; they remain stored on the server but remain accessible from the client. It can be argued that POP3 is more secure to cyber-attack because emails are locally stored. However, the server will be backed up regularly whereas the local client might not be. The major advantage for IMAP is that the server can be accessed from any client. This makes it suitable for anyone on the move or if someone is using a system other than that normally used. POP3 emails are only accessible from the one client system.

SMTP has been largely replaced by the use of web-based mail. A browser is used to access the email application, so HTTP is now the protocol used. However, SMTP remains in use for transfer between the mail servers.

Question 17.02

Which protocols are being used in your email system? Does your email system give you any choices?

FTP (File Transfer Protocol)

For routine transfers of files from one user to another the most likely method is to attach the file to an email. However, this is not always a suitable method. FTP (File Transfer Protocol) is the application-layer protocol that can handle any file transfer between two end-systems. File transfer is not simple if the end-systems have different operating systems with different file systems. FTP handles this by separating the control process from the data-transfer process.

Question 17.03

If you were setting up a website, you would need to upload files to a server. Do you know how you can use FTP for this from the system you normally use?

17.07 Peer-to-peer (P2P) file sharing

The network traffic generated by peer-to-peer (P2P) file sharing is one of the main features of Internet use. P2P is an architecture that has no structure and no controlling mechanism. Peers act as both clients and servers and each peer is just one end-system. When a peer acts as a server it is called a 'seed'.

The BitTorrent protocol is the most used protocol because it allows fast sharing of files. There are three basic problems to solve if end-systems are using BitTorrent.

1 How does a peer find others that have the wanted content?

   Every content provider should provide a content description, called a torrent, which is a file that contains the name of the tracker (a server that leads peers to the content) and a list of the chunks that make up the content. The torrent file is at least three orders of magnitude smaller than the content so can be transferred quickly. The tracker is a server that maintains a list of all the other peers (the 'swarm') actively downloading and uploading the content.

2 How do peers replicate content to provide high-speed downloads for everyone?

   Peers download and upload chunks at the same time, but peers have to exchange lists of chunks and aim to download rare chunks for preference. Each time a rare chunk is downloaded it automatically becomes less rare!

3 How do peers encourage other peers to provide content rather just using the protocol to download for themselves?

   This requires dealing with the free-riders or 'leechers' who only download. The solution is for a peer to initially randomly try other peers but then to only continue to upload to those peers that provide regular downloads. If a peer is not downloading or only downloading slowly, the peer will eventually be isolated or 'choked'.

It is worth noting that the language of BitTorrent is quite unusual and there are other terms used which have not been mentioned. Fortunately, the principles are straightforward.

Question 17.04

Who runs the tracker server? Are there alternative approaches for BitTorrent?

Reflection Point:

Networking is a large subject area with many interconnected concepts. Have you considered how you might make some structured notes to help with revision later?

Summary

- Circuit switching requires a dedicated circuit to be established between sender and receiver before transmission can start.
- In packet switching, packets can be transmitted without any circuit being created.
- Packet switching can use connectionless or connection-oriented transmission.
- A protocol suite is implemented as a layered stack.
- The TCP/IP protocol suite supports usage of the Internet.
- Ethernet is the most likely option for use in the Network and Data link layers.
- Examples of application-layer protocols are HTTP, SMTP, POP3, IMAP and FTP.
- Peer-to-peer file sharing on the Internet uses the BitTorrent protocol.

Exam-style Questions

1 The following represents the structure of an IP datagram.

IP Header	TCP Header	TCP Content

Using the code IP for IP Header, TCP for TCP Header and Content for TCP Content identify where the following data will be found:

a Destination address
b Sender address
c Destination port
d Sender port
e Packet sequence number
f Acknowledgement. [6]

2 Ethernet can be used in conjunction with the TCP/IP protocol suite.

a Draw a diagram to illustrate how the combination of Ethernet and the TCP/IP suite provides support for data communication. [5]

b Explain the meaning of the term 'MAC address'. [3]

3 One end-system with an Internet connection has a file. A user on another end-system connected to the Internet needs a copy of the file. There are different methods that might be used to enable the user to obtain a copy of the file.

a Identify **three** possible methods with a brief explanation for each. [6]

b Identify the application-layer protocols that each method will use with a brief explanation for each one. [8]

4 For sending and receiving emails the following application protocols might be used:

SMTP

POP3

IMAP

HTTP

For each of these protocols give a brief explanation as to how they might be used in association with an email application. You might find it useful to include a diagram for some of your account. [12]

5 **a** Four descriptions and three protocols are shown below.

Draw a line to connect each description to the appropriate protocol.

Description	Protocol used
email client downloads an email from an email server	HTTP
email is transferred from one email server to another email server	POP3
email client sends email to email server	SMTP
browser sends a request for a web page to a web server	

[4]

b Downloading a file can use the client-server model. Alternatively, a file can be downloaded using the BitTorrent protocol.

Name the model used. [1]

c For the BitTorrent protocol, explain the function of each of the following:

i tracker [2]

ii seed [2]

iii swarm [2]

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6 An email is sent from one email server to another using packet switching.

a State **two items** that are contained in an email packet apart from the data. [2]

b Explain the role of routers in sending an email from one email server to another. [3]

c Sending an email message is an appropriate use of packet switching.

Explain why this is the case. [2]

d Packet switching is not always an appropriate solution.

Name an alternative communication method of transferring data in a digital network. [1]

e Name an application for which the method identified in **part (d)** is an appropriate solution. Justify your choice. [3]

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